Autonomous Navigation of a Surface Vehicle in Waterways Using Artificial Vision

Oscar Eduardo Correa-Salazar[1], Agnelo Serrano-Almeida[1], Filiberto Muñoz[1,2]

[1]DIEM-SANAS, CINVESTAV, [2]IxM, SECIHTI

Background

1. Simple and robust method for water segmentation and detection of watercourse banks, utilizing vanishing point geometry and edge segmentation with YOLO v8.
2. Autonomous navigation in difficult-to-access environments.

Motivations

1. Facilitating water quality measurements with an autonomous surface vehicle.
2. Autonomous navigation with artificial vision.
3. Development of an autonomous surface vehicle.



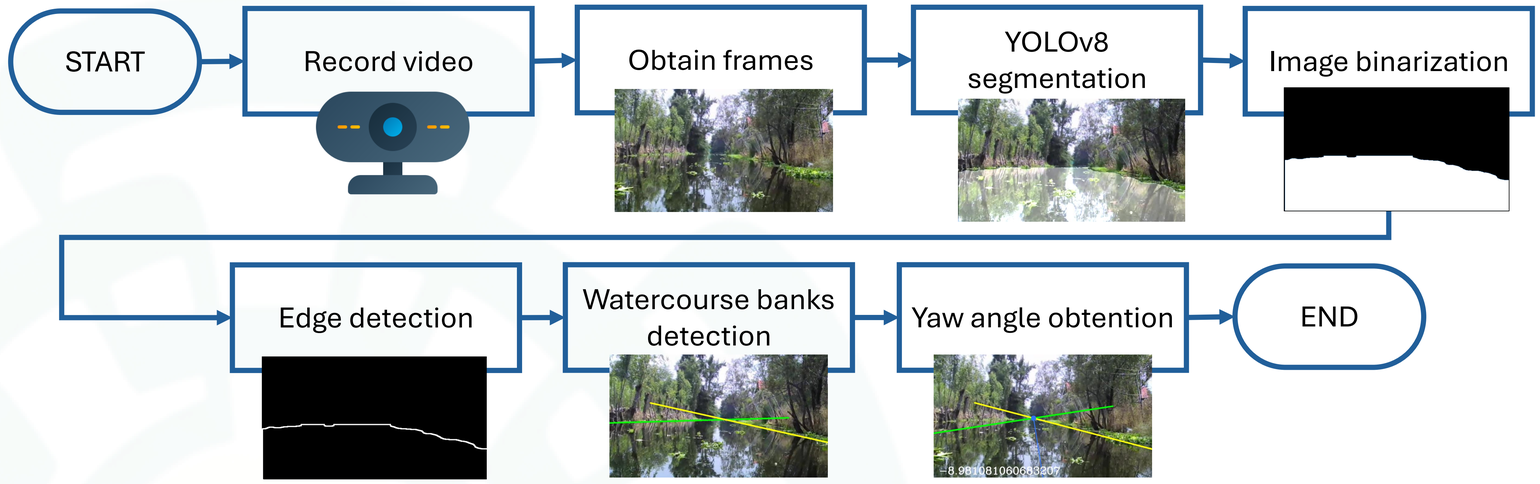


Higlights

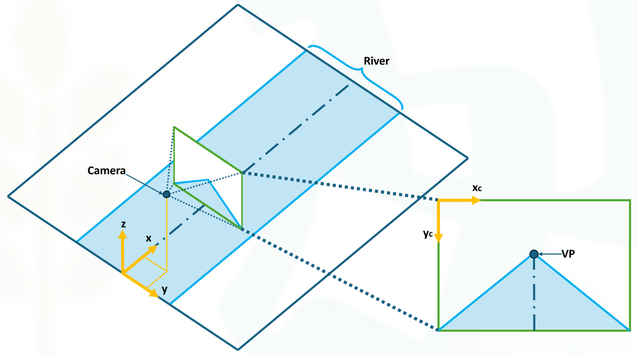
1. Autonomous navigation without GPS.
2. Segmentation with YOLO v8, a robust state-of-the-art and open-source model for artificial vision.

Video

1. Proposed method: Detection of watercourse banks using artificial vision and YOLOv8



2. Proposed method: Yaw angle detection using a vanishing point





3. Autonomous Navigation Results

